

REPUBLIQUE DU CAMEROUN

Paix-Travail-Patrie

MINISTRE DE L'ENSEIGNEMENT SUPERIEUR

COMMISSION NATIONALE D'ORGANISATION  
DES EXAMENS

REPUBLIC OF CAMEROON

Peace-Work-Fatherland

MINISTRY OF HIGHER EDUCATION

NATIONAL COMMISSION FOR THE ORGANISATION  
OF THE EXAMS

**NATIONAL EXAM OF HIGHER NATIONAL DIPLOMA- 2023 SESSION**

**Specialty/Option:** *Software Engineering*

**Paper:** *Practice of Computer*

**Duration:** 5 hours

**Credit Value:** 08

**SECTION A: ALGORITHM AND PROGRAMMING**

**/50 marks**

**I- C programming implementing a Simple Temperature Conversion Program 25 MARKS**

You have been tasked with implementing a simple temperature conversion program using C programming. The program should prompt the user to enter a temperature in Fahrenheit and convert it to Celsius. The program should output the result to the console.

1. What is a flowchart in programming? List its components (5 marks)
2. Draw the flowchart corresponding to the scenario (6 marks)
3. To implement this program, you will need to:
  - a. Implement user input. Your program should prompt the user for the temperature in Fahrenheit and read in the user's input. (3 marks)
  - b. Implement temperature conversion. Your program should use the formula  $(F - 32) * 5/9$  to convert the temperature from Fahrenheit to Celsius. (3 marks)
  - c. Implement error handling. Your program should handle errors such as invalid input (e.g., non-numeric values for the temperature). (5 marks)
  - d. Output the result. Your program should output the converted temperature to the console. (3 marks)

Your task is to implement a working prototype of this program, including the four components listed above. During the practical exam, you will be evaluated based on your ability to write clean, efficient, and well-documented C code that implements the temperature conversion logic correctly. You will also be evaluated on your ability to handle errors and debug your code to ensure that it works as expected.

**II- Object oriented programming**

**25 MARKS**

**Implementing a Simple Library Management System using Object-Oriented Programming\**



You have been tasked with implementing a simple library management system using object-oriented programming principles. The program should allow the user to add new books, remove books, and display a list of all books in the library. Each book should have a unique identifier, title, author, and number of copies available.

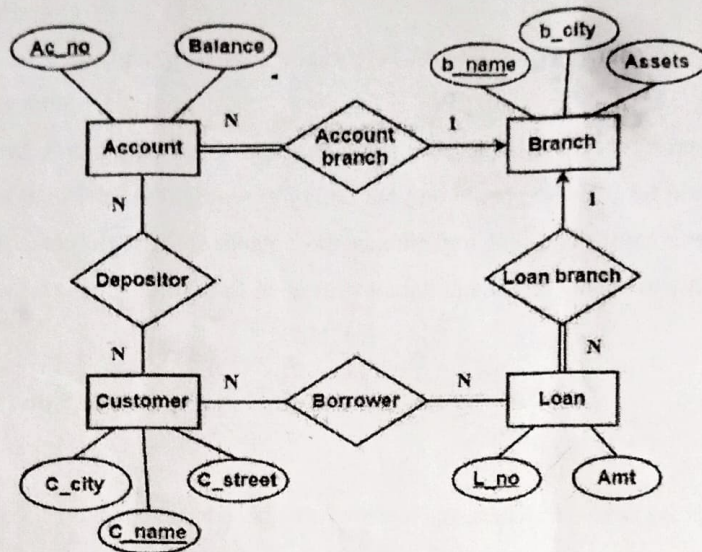
1. What is a class? Differentiate it from an object (4 marks)
2. To implement this program, you will need to:
  - a. Define a class for the book. Your class should have member variables for the book identifier, title, author, and number of copies available. (4 marks)
  - b. Implement constructors and getters/setters. Your class should have a constructor to initialize the member variables and getters/setters to access and modify the member variables. (5 marks)
  - c. Implement methods to add and remove books. Your program should allow the user to add new books to the library and remove existing books from the library. (4 marks)
  - d. Implement a method to display all books. Your program should allow the user to display a list of all books in the library. (4 marks)
  - e. Implement error handling. Your program should handle errors such as attempts to remove non-existent books. (4 marks)

Your task is to implement a working prototype of this program, including the five components listed above. During the practical exam, you will be evaluated based on your ability to write clean, efficient, and well-documented object-oriented code that implements the library management system logic correctly. You will also be evaluated on your ability to handle errors and debug your code to ensure that it works as expected.

## **SECTION B: DATABASE DEVELOPMENT AND ADMINISTRATION ( MYSQL) /20 marks**

To design an information system of a bank we have the following diagram:





- 1- What is this representing? Describe it with all his components (4 marks)
- 2- Give the equivalent relational model of this figure (5 marks)
- 3- Using appropriate software to create a database entitled BANK and create all the tables and populate each with 3 data's (8 marks)
- 4- Write a query to select all customer having 3 accounts and which have borrow 1500000 FCFA in your database and you screenshot it and save with the name HNDSWE2022 in a folder HND2023 (3 marks)

### SECTION C: WEB DESIGN

/15 marks

1. In others to design and executes a website, what do we need? (1.5 marks)
2. Differentiate between static and dynamic website (1.5 marks)
3. Implementing a Simple Contact Form using HTML, CSS, and JavaScript

You have been tasked with implementing a simple contact form using HTML, CSS, and JavaScript. The contact form should allow the user to input their name, email address, and message, and submit the form.

To implement this program, you will need to:

- a. Design the user interface. Your contact form should have a clean and user-friendly interface that displays the input fields, allows the user to input their name, email address, and message, and provides a submit button. (3 marks)
- b. Implement the contact form logic using JavaScript. Your contact form should use JavaScript to validate the input fields and send the form data to a server. (3 marks)



- c. Implement error handling. Your program should handle errors such as attempts to submit empty fields or invalid email addresses. (3 marks)
- d. Style the contact form using CSS. Your program should be visually appealing and use CSS to style the UI. (3 marks)

Your task is to implement a working prototype of this program, including the four components listed above. During the practical exam, you will be evaluated based on your ability to write clean, efficient, and well-documented HTML, CSS, and JavaScript code that implements the contact form logic correctly. You will also be evaluated on your ability to handle errors and debug your code to ensure that it works as expected.

#### **SECTION D: NETWORKING**

**/15 marks**

##### **Scenario:**

A small business called "Cafe Connect" has three computers and a printer that need to be connected to a network. The business owner wants to be able to share files between computers and access the printer from any computer on the network.

##### **Tasks:**

1. List the various devices which can be used to setup the network
2. Design a network diagram for Cafe Connect that includes the following devices: switch and three computers.
3. Configure the devices in the network diagram to provide the following functions:
  - Assign IP addresses to all devices in the network.
  - Configure the switch to provide connectivity between all devices in the network.
  - Share files between the three computers.
  - Enable the printer to be accessible from any computer on the network.
3. Test the network to ensure it is functioning properly.
4. Document the network diagram and device configurations.

##### **Time allocation:**

- Network design: 2 marks
- Device configuration: 5 marks
- Network testing: 2 marks
- Documentation: 1 mark



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HIGHER NATIONAL DIPLOMA EXAM (HND) EXAM

**National Exam of Higher National Diploma 2023 Session**

Speciality/Option: SOFTWARE ENGINEERING (SWE)

Paper : CASE STUDY

Duration : 6 hours

Credit : 14

**ANSWER ALL QUESTIONS**

**SECTION A: ALGORITHM AND PROGRAMMING (50 marks)**

**Question 1 Algorithms (10 marks)**

Consider the following loop:

Declare Number As Integer

Set Number = 1

Repeat

Write 2 \* Number

Set Number = Number + 1

Until Number > 10

Work required:

- Is this loop a pre-test loop or a post-test loop? Why?
- List the statements in the body of the loop.
- What is the test condition for this loop?
- What is the output of this loop?
- Draw the flowchart of this loop

2 marks

2 marks

2 marks

2 marks

2 marks

**Question 2 Pseudocode (25 marks)**

Consider the following pseudocode

Declare Q[10][10] As Integer

Declare R, C, As Integer

For (R = 1; R <= 3; R++)

For (C = 1; C <= 3; C++)

If R == C Then

Set Q[R][C] = 1

Else

Set Q[R][C] = 0



```

End If
End For(C)
End For(R)
For (R = 1; R <= 3; R++)
    For (C = 1; C <= 3; C++)
        Write Q[R][C]
    End For(C)
End For(R)

```

**Work required:**

- What is the output of code corresponding to the following pseudocode? **10 marks**
- Implement the previous pseudocode in the language you master most. **15 marks**

**Question 3 Object oriented programming (15 marks)**

- What are the two major components of an object? **(2 marks)**
- What is the relationship between a class and objects in that class? **(2 marks)**
- What is the difference between public and private class members? **(2 marks)**
- What is data hiding? Give an example of why it is important in an OOP program? **(2 marks)**  
*This is securing data in the class*
- Define a class called HND, including a constructor, that has one attribute named Value (of type Float) and two methods as follows:
  - SetValue() is a subprogram that imports a value of the attribute Value from the main program. **(2 marks)**
  - GetValue() is a function that returns the value of the attribute Value to the main program. **(2 marks)**
- What is a constructor? Why are constructors used? **(3 marks)**

**SECTION B: DATABASE DEVELOPMENT AND ADMINISTRATION (20 marks)**

**Question 4 (7 marks)**

- What is meant by normalization in database? **(1mark)**
  - State and explain the different norm form. **(3marks)**
  - For the figure below state weather it is normalize or not. If not, state the norm form and normalize the table. **(3marks)**



| DEPARTMENT     |                |           |                                |
|----------------|----------------|-----------|--------------------------------|
| DNAME          | <u>DNUMBER</u> | DMGRSSN   | DLOCATIONS                     |
| Research       | 5              | 333445555 | {Bellaire, Sugarland, Houston} |
| Administration | 4              | 987654321 | {Stafford}                     |
| Headquarters   | 1              | 888665555 | {Houston}                      |

**Question 5 (13 marks)**

- What is functional dependency (1mark)
- Give examples of DDL, DCL and DML as used in SQL (2marks)
- The questions below concerns the table below
  - What is the cardinality and degree of the relation (2marks)
  - List the attributes and choose a suitable primary key attribute for the relation. (2marks)
  - Write an SQL command to create the table below. The table name should be *student\_info* (1mark)
  - Write an SQL command to select all students with gpa greater than 2.7 (1.5marks)
  - Write an SQL command to select names of all the students (1mark)
  - Write an SQL command to add a fifth record to the table with *ID ST002, Name John, gpa 3.0* (1.5marks)
  - Write an SQL command to change the gpa of Suzy from 3.2 to 3.4 (1mark)

**SECTION C: WEB DESIGN (15 marks)**

**Question 6**

Write a HTML code to generate following output: (15 marks)



| Country | Population (In Crores) |      |
|---------|------------------------|------|
|         | 1998                   | 1999 |
| INDIA   | 85                     | 90   |
|         | 1999                   | 100  |
|         | 2000                   |      |
| USA     | 30                     | 35   |
|         | 1999                   | 40   |
|         | 2000                   |      |
| UK      | 25                     | 30   |
|         | 1999                   | 35   |
|         | 2000                   |      |

#### **SECTION D: NETWORKING (15 marks)**

##### **Question 7**

You have been provided with 2 computers and one computer that will act as the central computer to network and ensure that they should communication and share resources from a central computer.

##### **work required**

- What is a network? (1 mark)
- What is a computer network? (2 marks)
- What is the most appropriate OSs you think can be installed in both computers and the central computer so as to ensure that proper file sharing and resource access should be ensured? (2 + 1 = 3 marks)
- List all the various networking tools you will need in course of this activity to ensure that these two computers communicate with the central machine. (3 marks)
- Which other name can be given to the central machine? (2 marks)
- Which other name can be given to the two computers connected to the central machine? (1 mark)
- Which stand will you use to prepare your cables and why? (1 mark)
- LIST all the necessary configurations you think should be put in place and the various ping commands to ensure that the computers are on a network and can communicate and share resources (2 marks)



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**National Exam of Higher National Diploma– 2023 Session**

**Spécialty/option :** SWE. CSN. DBM

**Paper :** Digital Electronics

**Duration :** 4hours

**Credit :** 7/74

***Instructions:*** Answer all questions. Calculators of any type are not authorized!

**SECTION A: NUMBER SYSTEM AND CODES (20marks).**

**MCQs, for each question, four answers are proposed. Write the letter for the correct answer against the question number. Each question carries 1mark.**

- 1) The value of radix in binary number system is \_\_\_\_\_
  - a) 2
  - b) 8
  - c) 10
  - d) 1
- 2) Which of the following is not a binary number?
  - a) 1111
  - b) 101
  - c) 11E
  - d) 000
- 3) Positive integers must be represented by
  - a) signed numbers
  - b) unsigned number
  - c) Both A and B
  - d) None of above
- 4) An unweight code in which only one bit changes from one code number to the next is:
  - a) BCD
  - b) Excess-3
  - c) Gray
  - d) ASCII



- 5) In the decimal numbering system, what is the MSD?
- a) The middle digit of a stream of numbers
  - b) The rightmost digit to the decimal point of the number ✓
  - c) The last digit on the right ✓
  - d) The digit with the smallest weight
- 6) The following BCD number: 0010 1000.0111 0100 corresponds to the decimal number:
- a) 28.72 ✓
  - b) 29.52
  - c) 28.70 ✓
  - d) 28.82 ✓
- 7) How many binary digits are required to count to  $100_{10}$ ?
- a) 100
  - b) 3
  - c) 7
  - d) 2
- 8) Which of the following logical operations is represented by the + sign in Boolean algebra?
- a) Inversion
  - b) AND
  - c) OR
  - d) complementation
- 9) The binary equivalent of gray code 1111 is :
- a)  $(1101)_2$
  - b)  $(1011)_2$
  - c)  $(1110)_2$
  - d)  $(1010)_2$
10. The BCD number for decimal 347 is \_\_\_\_\_.
- a) 1100 1011 1000
  - b) 0011 0100 0111
  - c) 0011 0100 0001
  - d) 1100 1011 0110



11. Convert binary to octal:  $(110110001010)_2 = ?$
- a)  $(5512)_8$
  - b)  $(6612)_8$
  - c)  $(4532)_8$
  - d)  $(6745)_8$
12. What is the addition of the binary numbers 11011011010 and 010100101?
- a) 0111001000
  - b) 1100110110
  - c) 1110111111
  - d) 10011010011
13. Binary subtraction of  $100101 - 011110$  is
- a) 000111
  - b) 111000
  - c) 010101
  - d) 101010
14. Perform multiplication of the binary numbers:  $01001 \times 01011$
- a) 001100011
  - b) 110011100
  - c) 010100110
  - d) 101010111
15. Divide the binary numbers:  $1010001 \div 1000$
- a) 1010.001
  - b) 1000.001
  - c) 1100.001
  - d) 0011.001
16. Which numbering system uses numbers and letters as symbol?
- a) Decimal
  - b) binary
  - c) Octal
  - d) Hexadecimal



17. The octal number  $(651.124)_8$  is equivalent to \_\_\_\_\_

- a)  $(1A9.2A)_{16}$
- b)  $(1B0.10)_{16}$
- c)  $(1A8.A3)_{16}$
- d)  $(1B0.B0)_{16}$

18. The sign of a negative binary number is given by by its \_\_\_\_\_

- a) MSB
- b) LSB
- c) Byte
- d) Nibble

19. Which of the following is a correction detection code?

- a) Gray code
- b) Hamming code
- c) UNICODE
- d) Parity code

20. The format used to present the logic output for the various combinations of a logic expression is called:

- a) Boolean constant
- b) Boolean variable
- c) Truth table
- d) Logic function

## **SECTION B: COMPUTER FUNDAMENTAL (80marks)**

### **I) Hardware (20 Marks)**

1) Define the following acronyms:

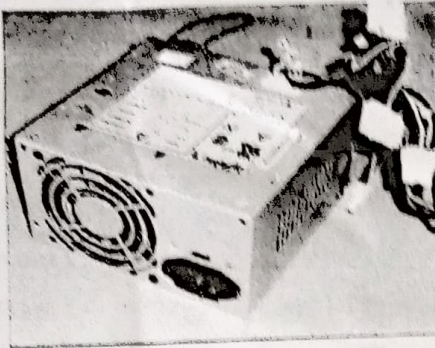
- a) RAM
- b) SDRAM
- c) EEPROM
- d) ROM
- e) DRAM

(5×1 = marks)



2) Name the following devices and give their roles

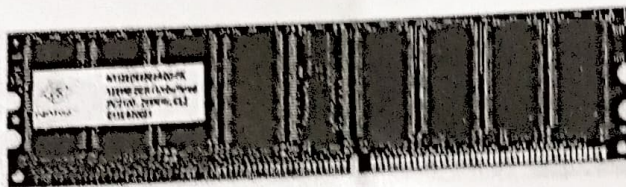
a)



b)



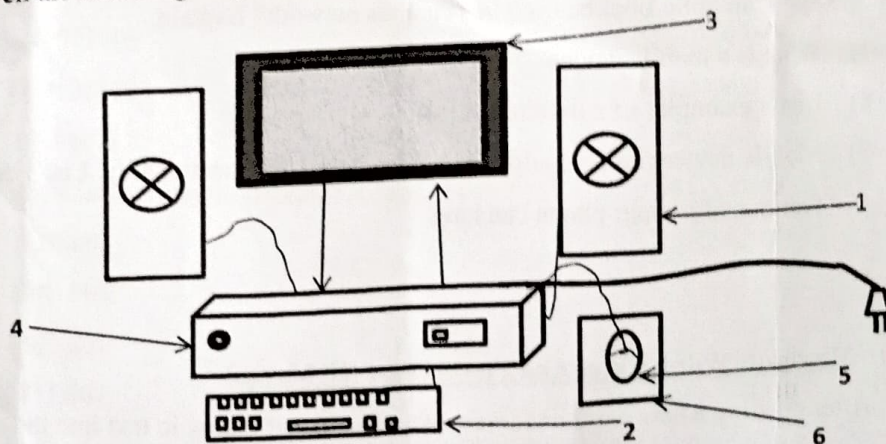
c)



(3 × 2 = 6 marks)

3) What is the main difference between a mainframe computer and a supercomputer?  
(2marks)

4) Given the following diagram below:



(3marks)

a) 4. a) Identify the various components from 1 to 6.

(4marks)

b) 4. b) Identify two input devices and two output devices.



## **II) Network And Mobile Devices (40 Marks)**

- 1) Give a brief description of the each of the following network devices
  - a) Router
  - b) Switch
  - c) Access point
  - d) Firewall
  - e) Repeater

(5 x 2 marks)
- 2) What is the Internet? 

(2 marks)
- 3) What is the difference between Internet and internet? 

(2 marks)
- 4) A company might decide to invest in an intranet. What are the benefits of this type of network? 

(2 marks)
- 5) How is Internet different from intranet? 

(2 marks)
- 6) Network can be classified using various criteria.
  - a) Classify networks based on geographical distances. 

(5 marks)
  - b) Classify networks based on the media of transmission. 

(3 marks)
  - c) What network type is suitable for a single school campus? 

(1 mark)
  - d) What transmission media is preferable for the network in (c) above. 

(1 marks)
  - e) Can optic fiber be used in a campus network? Explain. 

(2 marks)
- 7) What is a mobile device? 

(2 marks)
- 8) List 3 examples of a mobile device. 

(3 marks)
- 9) Mobile devices come equipped with hardware for functionality. List 5 common features of a smart phone can have 

(5 marks)

## **III) Microsoft Word, Excel And PowerPoint (20 Marks)**

- 1) After creating a new word document, a user can start typing in text into the new work space
  - a) What is the default tab of a new document? 

(1 mark)
  - b) What file extension does a word document save in? 

(1 mark)
  - c) Which tab is used to add illustrations to a word document 

(1 mark)
  - d) What is the role of the format painter in a Word document? 

(2 marks)



2) Consider the following figure below.

| 2  |   | CA Results | Exam Results | Total |
|----|---|------------|--------------|-------|
| 3  | 1 | 15         | 48           |       |
| 4  | 2 | 16         | 39           |       |
| 5  | 3 | 8          | 36           |       |
| 6  | 4 | 22         | 57           |       |
| 7  | 5 | 17         | 43           |       |
| 8  | 6 | 26         | 60           |       |
| 9  |   |            |              |       |
| 10 |   |            |              |       |

- a) Which Microsoft packages can be used to display the above table? (2 marks)
  - b) Which of the packages listed in (a) above contains dynamic squares? (1 mark)
  - c) What is the name given to each square? (1 mark)
  - d) What is the default data type of each dynamic box? (1 mark)
  - e) Suppose a teacher wants to obtain the totals of the students' scores, what steps are taken to quickly do this? (5 marks)
- 3) Graphic designers might use Microsoft office tools for productivity
- a) Which Microsoft tools can be used to design a flier (3 marks)
  - b) What file format does a PowerPoint file save in? (2 mark)



National Exam of Higher National Diploma 2023 Session

Spécialty/option: NWS, SWE, EDM  
Paper : DISCRETE MATHEMATICS  
Duration : 4 hours

Credits : 4/4/10

Instructions to candidates: Answer ALL questions

**SECTION A: MCQS – (20 marks)**

- 1) A ballot box contains 6 blue balls, 4 red balls and 2 green balls indiscernible on touch. One picks at random and simultaneously picks 3 balls. The probability not to obtain 3 balls of the same colour is:
- a)  $\frac{6^3+4^3}{12^3}$   
b)  $\frac{49}{55}$   
c)  $\frac{6}{55}$   
d)  $\frac{6 \times 5 \times 4 + 4 \times 3}{220}$
- 2) The code to access a residential building has 4 digits to be dialled on a numeric keypad with 10 digits from 0 to 9. We know that the code has exactly two digits of << 5 >>. The number of possible codes is:
- a)  $9^2$   
b)  $2 \times 9^2$   
c)  $3 \times 9^2$   
d)  $6 \times 9^2$
- 3) A player throws a biased die having six faces ones. He gains 10 points if 1 appears and 1 point if 2 or 4 appear. He gains no points if other number appears. Given that X is the Radom Variable for the number of points gained by the payer. The variance of X is :
- a) 2  
b) 13  
c) 16  
d) 17



4) The correlation between two quantitative variable X and Y is strong if the coefficient of correlation r is such that:

- a)  $|r| \geq 0.87$ ;
- b)  $r$  approximates to  $-1$ ;
- c)  $r$  approximates to  $0$ ;
- d)  $r$  approximates to  $1.5$

5) The value of:  $\lim_{x \rightarrow +\infty} 2e^{-2x}$  is:

- a)  $0$
- b)  $1$
- c)  $2$
- d)  $+\infty$

6) Let  $I_n = \int_0^1 x^n e^{-x} dx$ ; where n is natural number. Then.  $\lim_{n \rightarrow +\infty} I_n$  is equal to:

- a)  $+\infty$
- b)  $-\infty$
- c)  $0$
- d)  $1$

7) Let us consider the function  $f(x) = \frac{e^{2x}-1}{2e^x}$ . Then:

- a)  $f^{-1}(x) = \ln(x - \sqrt{x^2 + 1})$
- a.  $f^{-1}(x) = \ln(x + \sqrt{x^2 - 1})$
- b.  $f^{-1}(x) = \ln(x + \sqrt{x^2 + 1})$
- c.  $f^{-1}(x) = \ln(x - \sqrt{x^2 - 1})$

8) Let  $X_1, X_2, X_3$  be uniform random variables on the interval  $(0, 1)$  with  $\text{Cov}(X_i, X_j) = 1/24$  for  $i, j = 1, 2, 3, i \neq j$ . Calculate the variance of  $X_1 + 2X_2 - X_3$ .

- a)  $1/6$
- b)  $5/12$
- c)  $1/4$
- d)  $1/2$



9) Given that  $X \sim B(n, 0.3)$  and  $E(X) = 4.6$ , the value of  $n$  is

- a) 12.80
- b) 15.33
- c) 4.90
- d) 48.00

10) The probability distribution function of a continuous random variable  $X$  is

$$f(x) = \begin{cases} kx(2-x), & 0 \leq x \leq 2 \\ 0, & \text{else where} \end{cases}$$

The value of the constant  $K$  is

- a)  $\frac{4}{3}$
- b)  $\frac{3}{20}$
- c)  $\frac{3}{4}$
- d) 1

11) Let  $f$  be a continuous and strictly decreasing function defined in  $[a, b]$ ,  $a < b$ . Let

$f^{-1}(x) = g(x)$ . then

- a)  $g'(x) < 0$
- b)  $g'(x) > 0$
- c)  $g(x) < 0$  D.
- d)  $g(x) > 0$

12) Given a sequence  $(U_n)$  defined by  $U_0 = 0$  and  $\forall n \geq 1, U_n = \sqrt{U_{n-1} + 6}$ , where  $n$  is a natural number. The sequence  $(U_n)$  is

- a) Divergent
- b) Convergent
- c) Decreasing
- d) Constant

13) The integral  $I = \int 7^{3x} dx$  is equal to:

- a)  $I = 3(7^{3x})$
- b)  $I = \frac{7^{3x}}{3}$
- c)  $I = \frac{7^{3x} \ln 7}{3}$
- d)  $I = (3 \ln 7) 7^{3x}$



14) The value of  $\sum_{k=1}^n (\frac{1}{k} - \frac{1}{k+1})$  is;

- a)  $\frac{1}{n+1}$
- b)  $\frac{n}{n+1}$
- c)  $\frac{n+1}{n}$
- d)  $\frac{1}{n+1} - 1$

15) Let us consider  $\mathbf{u} = -2\mathbf{i} + \mathbf{j} + \mathbf{k}$ ,  $\mathbf{v} = \mathbf{i} - \mathbf{j} + \mathbf{k}$  and  $\mathbf{w} = \mathbf{i} - \mathbf{j} + 2\mathbf{k}$ . Then:

- a)  $\mathbf{v} \cdot \mathbf{w} = 0$
- b)  $\mathbf{u} \cdot \mathbf{v} = 2$
- c)  $\mathbf{w} \cdot \mathbf{u} = -1$
- d)  $\mathbf{w} \cdot \mathbf{v} = 5$

16) If  $y = (ax + 1)e^{2x}$  is a solution to the differential equation  $y'' - 2y' - y = (x - 1)e^{2x}$ , then  $a =$

- a) 2
- b)  $\frac{1}{2}$
- c) -1
- d) -2

17) The domain of the function  $f: \mathbb{R} \rightarrow \mathbb{R}$  defined by  $f(x) = \sqrt{\frac{x-1}{x}}$  is

- a)  $\{x: x < 0 \text{ and } x \geq 2\}$
- b)  $\{x: x < 0 \text{ or } x \geq 2\}$
- c)  $\{x: x < 0\}$
- d)  $\{x: x \geq 2\}$

18) A and B are two independent events with  $p(A) = 0.5$  and  $p(B) = 0.2$ . The probability of the event  $A \cup B$  is

- a) 0.1
- b) 0.7
- c) 0.6
- d) 0.8



19) In a shop, a tray contains marked down notebooks. We know that 50% notebooks are spiral-bound and 75% of notebook are large-squared. Among the large-squared notebooks, 40% are spiral-bound. Larissa randomly chooses a spiral-bound notebook. The probability that it is large squared is

- a) 0.3
- b) 0.5
- c) 0.6
- d) 0.75.

20) The function  $f: \mathbb{R} \rightarrow \mathbb{R}$  defined by  $f(x) = 2x^3 + x$

- a) A. periodic
- b) neither odd nor even
- c) even
- d) odd

21) For  $y = \sin^2 x + \cos^2 x$ ,  $\frac{dy}{dx} =$

- a)  $2 \sin x - 2 \cos x$
- b)  $2 \sin x \cos x$
- c)  $4 \sin x \cos x$
- d) 0

22) If  $y^3 = x^2$ , then  $\frac{dy}{dx} =$

- a)  $\frac{2x}{y^3}$
- b)  $\frac{2x}{3y^2}$
- c)  $\frac{x^2}{3y^2}$
- d)  $\frac{x^2}{3y}$



23)  $\csc\left(-\frac{4}{3}\pi\right) =$

a) 2

b)  $\frac{\sqrt{3}}{2}$

c)  $\frac{2\sqrt{3}}{3}$

d)  $-\frac{2\sqrt{3}}{3}$

24) Find a positive value  $c$ , for  $x$ , that satisfies the conclusion of the Mean Value Theorem for Derivatives for  $f(x) = 3x^2 - 5x + 1$  on the interval  $[2, 5]$ .

a) 1

b)  $\frac{7}{2}$

c) 3

d)  $\frac{23}{6}$

25) At what value of  $x$  does the function  $f(x) = \frac{(x+1)^2}{x^2 - 1}$  have a removable discontinuity?

a) 1

b) 3

c) 2

d) -1

26) Suppose  $g$  is a function defined on the interval  $[0, 5]$  such that  $\lim_{x \rightarrow 2} g(x) = -1$ , find

$\lim_{x \rightarrow 2} [g(x)]^3$

a) 8

b)  $\frac{1}{8}$

c) -1

d)  $\frac{1}{3}$



27) If  $f(a) = f(b) = 0$  and  $f(x)$  is continuous on  $[a, b]$ , then

- a)  $f(x)$  must be identically zero
- b)  $f'(x)$  may be different from zero for all  $x$  on  $(a, b)$
- c) There exists at least one number  $c$ ,  $a < c < b$ , such that  $f'(c) = 0$
- d)  $f'(x)$  must exist for every  $x$  on  $(a, b)$

28) A medical treatment has a success rate of .8. Two patients will be treated with this treatment. Assuming the results are independent for the two patients, what is the probability that neither one of them will be successfully cured?

- a) 0.5
- b) 0.36
- c) 0.2
- d) 0.04

29) Suppose that the probability of event A is 0.2 and the probability of event B is 0.4. Also, suppose that the two events are independent. Then  $P(A|B)$  is:

- a)  $P(A) = 0.2$
- b)  $P(A)/P(B) = 0.2/0.4 = \frac{1}{2}$
- c)  $P(A) \times P(B) = (0.2)(0.4) = 0.08$
- d) None of the above.

30) A specific range of numbers within which a population mean should lie is

- a) the range.
- b) the confidence coefficient.
- c) the confidence interval.
- d) the confidence level.

31) Calculate the sample standard deviation using the following data:

X: 4    5    2    6    4    3    4

The standard deviation is which of the following?

- a) 10
- b) 1.66
- c) 1.20
- d) 1

33) A sample of 40 cows is drawn to estimate the mean weight of a large herd of cattle. If the standard deviation of the sample is 96 kg, what is the maximum error in a 90% confidence interval estimate?

- a) 25 kg
- b) 158 kg
- c) 58 kg
- d) 30 kg

34) The Laplace transform of  $f(t) = 3$  is

- a)  $\frac{3}{s}$
- b)  $\frac{1}{s}$
- c) 3
- d) 3s

34) The inverse Laplace transform of **Erreur !** is

- a)  $1 - \text{Erreur !}t$
- b)  $1 - \text{Erreur !}t + \text{Erreur !}t$
- c)  $1 - \text{Erreur !}t + \text{Erreur !}t^2$
- d)  $\text{Erreur !}t + \text{Erreur !}t^2$

35) The eigenvalues of  $A = \begin{pmatrix} 1 & 3 \\ 4 & 2 \end{pmatrix}$  are.

- a) 5 and -2
- b) 2 and 5
- c) 3 and 1
- d) -3 and -1

36) The interval of convergence the function  $f(x) = x^2 \ln(1 - 2x)$

- a)  $I = (1, 2)$
- b)  $I = [-\frac{1}{2}, \frac{1}{2})$
- c)  $I = [1, 2]$
- d)  $I = [-\frac{1}{2}, \frac{1}{2}]$



37)  $f(x) = \sqrt{4+x}$  as a Maclaurin series has as its radius of convergence

- a)  $R = 2$
- b)  $R = 0$
- c)  $R = 3$
- d)  $R = 1$

38)  $\sec^2 x - \tan^2 x$  is an identity for which of the following:

- a) A. 1
- b)  $\sin^2 x - \cos^2 x$
- c)  $\cos^2 x - \sin^2 x$
- d)  $1 - \tan^2 x$

39) What is the magnitude of the vector  $6\mathbf{i} + 15\mathbf{j}$ ?

- a)  $\sqrt{21}$
- b) 261
- c) 21
- d)  $\sqrt{261}$

40) What are the first five terms of the sequence defined as

$$a_1 = 3$$

$$a_{n+1} = a_n - 4, \text{ for } n \geq 1?$$

- a)  $-3, -2, -1, 0, 1$
- b)  $-1, -5, -9, -13, -17$
- c)  $3, -1, -5, -9, -13$
- d)  $3, -1, 0, 1, 2$

## SECTION B: STRUCTURAL

### 1. ANALYSIS(30 marks)

1.1 State Rolle's Theorem. Find the value of  $x_0$  prescribe in Rolle's Theorem for

$f(x) = x^3 - 12x$  on the interval  $0 \leq x \leq 2\sqrt{3}$ . (5 marks)

1.2 Evaluate the double integral below. (5 marks)

$$\int_{-1}^2 \int_{2x^2-2}^{x^2+x} x \, dy \, dx.$$

1.3 If  $f(x, y) = x^3 e^{5y} + y \sin 2x$ .

(5 marks)

Find

$$f_x, f_y, f_{xx}, f_{yy}, f_{xy}$$



1.4 Find a)  $\nabla \cdot \vec{F}$  ( $\text{div}(\vec{F})$ ) b)  $\text{grad}(\text{div}(\vec{F}))$  and c)  $\nabla \times \vec{F}$  ( $\text{curl}(\vec{F})$ ) if

$$\vec{F} = (y + x^2)\vec{i} + z\vec{j} + x^2\vec{k} \quad (5 \text{ marks})$$

1.5 Use Laplace transform to solve the  $dy$  following differential equation:

$$\frac{d^2y}{dx^2} - 7\frac{dy}{dx} + 10y = e^{2x} + 20. \text{ Given that when } x = 0, y = 0, \frac{dy}{dx} = -\frac{1}{3} \quad (10 \text{ marks})$$

## 2. STATISTICS (30 marks)

2.1 An inquiry in the salary of workers of an enterprise is concerned with the sample of 58 workers. The results of the inquiry are shown in the table below:

|                    |             |             |             |             |             |             |             |
|--------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| Hourly Salary(CFA) | 1440 - 1620 | 1620 - 1800 | 1800 - 1980 | 1980 - 2160 | 2160 - 2340 | 2340 - 2520 | 2520 - 2700 |
| Number of workers  | X           | 2           | 4           | 8           | 11          | 15          | Y           |

- Knowing that the average hourly salary of 58 workers of the enterprise is 2302,76CFA, determine the missing frequency values, x and y. (9 Marks)
- Determine the median salary of the 58 workers of the enterprise. (4 Marks)
- Calculate the first and the third quartile. (3 Marks)

2.2 A Credit Union has a portfolio of four branches Edea (E), Garoua (G), Douala(D), Limbe (L).

These branches are expected to generate profits as follows:

| Branch          | Edea (E) | Garoua (G) | Douala(D) | Limbe (L) |
|-----------------|----------|------------|-----------|-----------|
| Profit (000frs) | 2000     | 3000       | 4500      | 6000      |

The likelihood that each branch will generate the expected profit is as follows:

| Branch      | Edea (E) | Garoua (G) | Douala(D) | Limbe (L) |
|-------------|----------|------------|-----------|-----------|
| Probability | 0.3      | 0.2        | 0.1       | 0.4       |



- a) Calculate the expected profit that will be generated by Credit Union. (6 Marks)
- b) Determine the variance of the profits. (6 Marks)
- c) Calculate the standard deviation of profits. (2 Marks)

### 3. PROBABILITY (20 marks)

#### 3.1

- a) The probability distribution of a discrete random variable  $X$  is given by

$$P(X = x) = \begin{cases} k(x + 1), & x = 1, 2, 3 \\ k(x - 3), & x = 4, 5, 6 \\ 0, & \text{elsewhere} \end{cases}$$

Find ;

- i. the value of the constant  $k$ .
- ii. the mean of  $X$
- iii. the variance of the distribution (3+2+4 = 9 Marks)

- 3.2 Of the large number of candidates applying for a job in an enterprise, 5% are qualified. The candidates are interviewed one after another until the first qualified applicant is interviewed.

- a) Find the mean number of applicants interviewed.
- b) Calculate, to 4 decimal places, the probability that the fifth candidate interviewed is the first qualified applicant. (2+3 = 4 marks)

- 3.3 Two machines  $X$  and  $Y$  produce, respectively, 60 % and 40 % of the total number of electrical fuses from a certain factory. The probabilities of faulty fuses produced by these machines are  $\frac{2}{25}$  and  $\frac{1}{20}$  respectively. A fuse is selected at random from a consignment of fuses from the factory. Find the probability that

- a) The fuse is faulty,
- b) The fuse is faulty or it comes from  $Y$
- c) The fuse comes from  $X$  given that it is faulty (2+ 2+ 3 = 7 Marks)



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Peace-Work-Fatherland

MINISTRY OF HIGHER EDUCATION

NATIONAL COMMISSION FOR THE ORGANISATION OF  
HIGHER NATIONAL DIPLOMA EXAM (HND) EXAM

### National Exam of Higher National Diploma 2023 Session

Speciality/Option: SOFTWARE ENGINEERING (SWE)

Paper : Information Systems

Duration : 3 hours

Credit : 6

### ANSWER ALL QUESTIONS

#### SECTION A: SYSTEM ARCHITECTURE (40 marks)

- 1) What is an information system in an organization? 4mks
- 2) Name the five important components of an information system? 4mks
- 3) Draw an diagram to show relation of information system to the levels of organization. 4mks
- 4) What are the functions of an information system? 4mks
- 5) Name the steps involved in the development of an information system. 4mks
- 6) Define the client server architecture concept. 4mks
- 7) What are the 3 functions of client-server? 4mks
- 8) What are the two broad classes of middleware in client server environment? 4mks
- 9) What are the five major technologies that can be used to create Client/Server applications in client server environment? 4mks
- 10) What is public cloud and private cloud? 4mks

#### SECTION B: OPERATING SYSTEMS (40 marks)

##### **I. MCQs (20 Marks)**

Each Correct Answer Carries One (01)

1. CPU fetches the instruction from memory according to the value of:
  - A. program counter
  - B. status register
  - C. instruction register
  - D. program status word



2. A memory buffer used to accommodate a speed differential is called:
  - A. stack pointer
  - B. cache
  - C. accumulator
  - D. disk buffer
3. Which one of the following is the address generated by CPU?
  - A. physical address
  - B. absolute address
  - C. logical address
  - D. none of the mentioned
4. Run time mapping from virtual to physical address is done by:
  - A. memory management unit
  - B. CPU
  - C. PCI
  - D. none of the mentioned
5. Memory management technique in which system stores and retrieves data from secondary storage for use in main memory is called:
  - A. fragmentation
  - B. paging
  - C. mapping
  - D. none of the mentioned
6. Effective access time is directly proportional to:
  - A. page-fault rate
  - B. hit ratio
  - C. memory access time
  - D. none of the mentioned
7. In FIFO page replacement algorithm, when a page must be replaced:
  - A. oldest page is chosen
  - B. newest page is chosen
  - C. random page is chosen
  - D. none of the mentioned

8. Which algorithm chooses the page that has not been used for the longest period of time whenever the page required to be replaced?
- A. first in first out algorithm
  - B. additional reference bit algorithm
  - C. least recently used algorithm
  - D. counting based page replacement algorithm
9. A process is thrashing if:
- A. it is spending more time paging than executing
  - B. it is spending less time paging than executing
  - C. page fault occurs
  - D. swapping can not take place
10. Working set model for page replacement is based on the assumption of:
- A. modularity
  - B. locality
  - C. globalization
  - D. random access
11. The operating system is :
- A. in the low memory
  - B. in the high memory
  - C. either a or b (depending on the location of interrupt vector)
  - A. D. None of these
12. In contiguous memory allocation :
- A. each process is contained in a single contiguous section of memory
  - B. all processes are contained in a single contiguous section of memory
  - C. the memory space is contiguous
  - D. None of these
13. The relocation register helps in :
- A. providing more address space to processes
  - B. a different address space to processes
  - C. to protect the address spaces of processes
  - A. D. None of these



14. With relocation and limit registers, each logical address must be \_\_\_\_\_ the limit register.
- A. less than
  - B. equal to
  - C. greater than
  - D. None of these
15. The operating system and the other processes are protected from being modified by an already running process because :
- A. they are in different memory spaces
  - B. they are in different logical addresses
  - C. they have a protection algorithm
  - D. every address generated by the CPU is being checked against the relocation and limit registers
16. Transient operating system code is code that :
- A. is not easily accessible
  - B. comes and goes as needed
  - C. stays in the memory always
  - D. never enters the memory space
17. Using transient code, \_\_\_\_\_ the size of the operating system during program execution.
- A. increases
  - B. decreases
  - C. changes
  - D. maintains
18. When memory is divided into several fixed sized partitions, each partition may contain \_\_\_\_.
- A. exactly one process
  - B. at least one process
  - C. multiple processes at once
  - D. None of these
19. In fixed sized partition, the degree of multiprogramming is bounded by \_\_\_\_\_.
- A. the number of partitions
  - B. the CPU utilization
  - C. the memory size
  - D. All of these

20. The first fit, best fit and worst fit are strategies to select a \_\_\_\_\_.

- A. process from a queue to put in memory
- B. processor to run the next process
- C. free hole from a set of available holes
- D. All of these

**II. Structural Questions (20 Marks)**

- 1. What does a systems analyst do? **4 marks**
- 2. Name the roles of a systems analyst. **4 marks**
- 3. What are the characteristics of a good system analyst? **4 marks**
- 4. What are the four skills of a systems analyst? **4 marks**
- 5. Name the four types of feasibility study. **4 marks**

**SECTION C: PROJECT MANAGEMENT AND LEGAL REGULATIONS (20 marks)**

1. Define the following terms:

- a) Project **1 mark**
- b) Software project manager **1 mark**

2. Software project management comprises of a number of activities, which contains planning of project, deciding scope of software product, estimation of cost in various terms, scheduling of tasks and events, and resource management.

- a) What are the activities that are carried out during project scope management activities? **4 marks**
- b) What is usually estimated during project estimation phase? **7 marks**
- c) Discuss two types of Project Estimation Techniques. **2 marks**

3.

- a) What is Project Scheduling? **2 marks**
- b) What are the activities that are carried out during Project Scheduling? **4 marks**



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### National Exam of Higher National Diploma 2023

Specialty/option : SWE, HWM, CWD, FSD

Paper : DIGITAL LITERACY

Duration : 2 HOURS

Credit :1

**Instructions:** Follow the instructions below carefully

- This paper contains two sections; Section A (50%) and Section B (50). Answer all questions in each section.
- The respective mark(s) of each question is indicated.
- It is a closed book exam and pre-prepared material is not authorized.
- Should any student encounter any difficulty, only the examiner(s) in the examination hall should be contacted

### **SECTION A: GENERAL COMPUTER KNOWLEDGE** (50 marks)

#### **I- MULTIPLE CHOICE QUESTIONS** (20 marks)

1. Which of the following is a temporal storage device?
  - A. Rom
  - B. RAM
  - C. Hard drive
  - D. Tape drive.
2. Which is considered the brain of a computer?
  - A. Motherboard
  - B. RAM
  - C. CPU
  - D. Power Supply.
3. Which keyboard command is used to launch the start menu?
  - A. ALT + ESC
  - B. CTRL + ESC
  - C. CTRL + TAB
  - D. SHIFT + ESC.

4. From which computer generation, did operating systems developed?
- A. First
  - B. Second
  - C. Third
  - D. Fourth
5. Which generation of computer is still under development
- A. Fourth Generation
  - B. Fifth Generation
  - C. Sixth Generation
  - D. Seventh Generation
6. Which one of the following software applications would be the most appropriate for performing numerical and statistical calculations?
- A. Database
  - B. Document processor
  - C. Graphics package
  - D. Spreadsheet
7. The primary different between a LAN and WAN is:
- A. Number of programs install
  - B. Variety of hardware devices
  - C. Distance
  - D. Number of hardware devices
8. Which of the following is designed to control the operation of the computer?
- A. System software
  - B. Application software
  - C. User
  - D. Device drivers
9. Every device that is attached to the computer is connected to \_\_\_\_\_ -
- A. Main memory
  - B. BIOS
  - C. Mother board
  - D. Monitor



**10. Which of these device is not an example of a computer peripheral?**

- A. Hard disk
- B. Mouse
- C. Keyboard
- D. RAM

**11. Is not an example of a multitask operating system**

- A. MS Access
- B. MS Windows
- C. MS DOS
- D. All of the above

**12. Digitization is the process of converting information into (1 mark)**

- A. an analog format
- B. the analog and digital formats
- C. a digital format
- D. a matlab format

**13. An example of low level language programing is:(1 mark)**

- A. Pascal
- B. Fortran
- C. Assembly language
- D. C+

**14. The language used by a computer is called: (1 mark)**

- A. english language
- B. french language
- C. binary language
- D. computer language

**15. Resources that can be managed by the operating system are: (1 mark)**

- A. hardware resources
- B. graphical user interface, hardware and software resources
- C. graphical user interface
- D. software resources

**16. What is digital literacy**

- A. A course that deals with computers and internet
- B. One's ability to comprehend and manipulate digital information
- C. The ability to write codes and produce software
- D. None of the above

**17. You create a presentation by using PowerPoint. Which of the following toolbars will you use to add geometric shapes to the presentation?**

- A. Formatting
- B. Drawings
- C. Illustrations
- D. Reviewing

**18. What is a computer network?**

- A. A connection of two computers
- B. An interconnection of devices
- C. A connection of computers for sharing resources
- D. A service provider such as MTN and Orange

**19. List two types of maintenance you know of**

- A. Protective and Corrective
- B. Preventive and Corrective
- C. Corrective and Pre-emptive
- D. Protective and Creative

**20. <http://www.google.com> is an example of a**

- A. WWW
- B. URL
- C. HTML
- D. Mozilla Firefox



## **II- Structural Questions (30marks)**

1. According to the generation of computers, explain the technology of the first generation computer. What device was used as Input at that time? (5 marks)
2. Give any two characteristics of the computer and explain them. (5 marks)
3. Differentiate between computer architecture and computer organization? (4 marks)
4. List 4 advantages of digital communication over analogue communication. (4 marks)
5. Computer systems are made of both hardware and software
  - a. List 10 hardware components of a PC. (5 marks)
  - b. How can software be classified. (2 marks)
6. Define the following terms as used in computing (3 marks)
  - a. Shareware
  - b. Freeware
  - c. Open source
7. List two differences between RAM and ROM? (2 marks)

## **SECTION B: WEB AND DIGITAL SOCIETY (50 marks)**

### **I- WEB KNOLEGDE ( 20 marks)**

1. Give the meaning of TCP, IP, HTTP, HTTPS, (2 marks)
2. Name 4 internet search engines. (4 marks)
3. Name 4 internet browsers (4 marks)
4. Differentiate between internet search engines and internet browsers (2 marks)
5. What is internet of things? (2 marks)
6. Differentiate between the Web 1.0 and the Web2.0 (2 marks)
7. Differentiate between the Web 3.0 and the Web 4.0? (2 marks)
8. Do you thing that e-banking is helpful for us? (2 marks)

### **II- Digital society (30 marks)**

1. Give 2 advantages and 2 disadvantages of using social media. (2x2=4 marks)
2. Give two benefits each of E-health and E-banking? (2x2=4 marks)

3. What do you understand by fake news? and list two consequences of propagating fake news **(2x2 =4marks)**
4. Name the three services delivered by the cloud computing companies. **(3x1 marks)**
5. Explain the following:
- Software piracy. **(2 marks)**
  - Media literacy. **(2 marks)**
6. Explain what is digital communication and (b)state its 3 sources. **(5 marks)**
7. Explain the following as used in emailing: **(6 marks)**
- a. Spamming
  - b. Phishing
  - c. Attachments



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**National Exam of Higher National Diploma 2023 Session**

**Speciality/Option: SOFTWARE ENGINEERING (SWE)**

**Paper : Systems Analysis and Design**

**Duration : 4 hours**

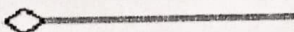
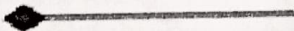
**Credit : 7**

**ANSWER ALL QUESTIONS**

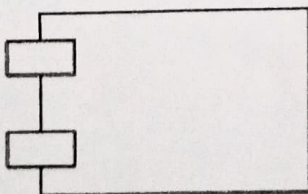
**SECTION A: OBJECT MODEL (25 marks)**

**MCQs each correct answer carries one (01) mark**

1. Which of the following UML diagrams has a static view?
  - a) Collaboration
  - b) Use case
  - c) State chart
  - d) Activity
2. What type of core-relationship is represented by the symbol in the figure below?

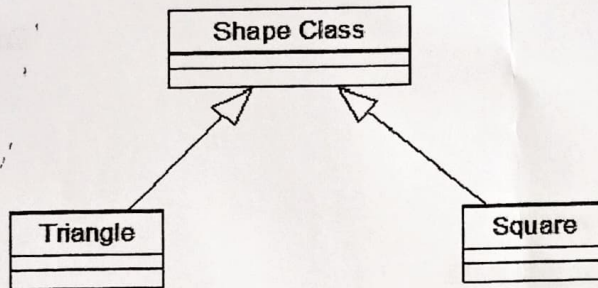


- a) Aggregation
  - b) Dependency
  - c) Generalization
  - d) Association
3. Which core element of UML is being shown in the figure?



- a) Node
- b) Interface
- c) Class
- d) Component

4. What type of relationship is represented by Shape class and Square ?



- a) Realization
- b) Generalization
- c) Aggregation
- d) Dependency

5. Which diagram in UML shows a complete or partial view of the structure of a modeled system at a specific time?

- a) Sequence Diagram
- b) Collaboration Diagram
- c) Class Diagram
- d) Object Diagram

6. Interaction Diagram is a combined term for

- a) Sequence Diagram + Collaboration Diagram
- b) Activity Diagram + State Chart Diagram
- c) Deployment Diagram + Collaboration Diagram
- d) None of the mentioned

7. Which of the following diagram is time oriented?

- a) Collaboration
- b) Sequence
- c) Activity
- d) None of the mentioned



8. How many diagrams are there in Unified Modelling Language?
- a) six
  - b) seven
  - c) eight
  - d) nine
9. Which of the following is not needed to develop a system design from concept to detailed object-oriented design?
- a) Designing system architecture
  - b) Developing design models
  - c) Specifying interfaces
  - d) Developing a debugging system
10. Which of the following is a dynamic model that shows how the system interacts with its environment as it is used?
- a) system context model
  - b) interaction model
  - c) environmental model
  - \*d) both system context and interaction
11. Which of the following is a structural model that demonstrates the other systems in the environment of the system being developed?
- a) system context model
  - b) interaction model
  - c) environmental model
  - \*d) both system context and interaction
12. Which of the following come under system control?
- a) Reconfigure
  - b) Shutdown
  - c) Powersave
  - \*d) All of the mentioned

13. Which model describes the static structure of the system using object classes and their relationships?
- a) Sequence model
  - b) Subsystem model
  - c) Dynamic model
  - d) Structural model
14. Which model shows the flow of object interactions?
- a) Sequence model
  - b) Subsystem model
  - c) Dynamic model
  - d) Both Sequence and Dynamic model
15. Which message is received so that the system moves to the Testing state, then the Transmitting state, before returning to the Running state?
- a) signalStatus()
  - b) remoteControl()
  - c) reconfigure()
  - d) reportStatus()
16. A description of each function presented in the DFD is contained in a \_\_\_\_\_
- a) data flow
  - b) process specification
  - c) control specification
  - d) data store
17. Which diagram indicates the behaviour of the system as a consequence of external events?
- a) data flow diagram
  - b) state transition diagram
  - c) control specification diagram
  - d) workflow diagram
18. A data model contains
- a) data object
  - b) attributes
  - c) relationships
  - d) all of the mentioned



19. \_\_\_\_\_ defines the properties of a data object and take on one of the three different characteristics.
- a) data object
  - b) attributes
  - c) relationships
  - d) data object and attributes
20. The \_\_\_\_\_ of a relationship is 0 if there is no explicit need for the relationship to occur or the relationship is optional.
- a) modality
  - b) cardinality
  - c) entity
  - d) structured analysis
21. A \_\_\_\_\_ is a graphical representation that depicts information flow and the transforms that are applied as data moves from input to output.
- a) data flow diagram
  - b) state transition diagram
  - c) control specification
  - d) workflow diagram
22. . A data condition occurs whenever a data is passed to an input element followed by a processing element and the result in control output.
- a) True
  - b) False
23. The \_\_\_\_\_ enables the software engineer to develop models of the information domain and functional domain at the same time
- a) data flow diagram
  - b) state transition diagram
  - c) control specification
  - d) activity diagram
24. The \_\_\_\_\_ contains a state transition diagram that is a sequential specification of behavior.
- a) data flow diagram
  - b) state transition diagram
  - c) control specification
  - d) workflow diagram



25. Which of the following is not a construct?

- a) sequence
- b) condition
- c) repetition
- d) selection

**SECTION B: DATABASE-MYSQL (25marks)**

- 1) What are the three major steps of the database design (data modeling) process?  
Define each in *one* sentence. **2marks**
- 2) What types of participation constraints can you have in an E-R model? Define each by *one* sentence. **3marks**
- 3) What are ACID properties of transactions? Explain each by one sentence. **5marks**
- 4) **10marks**

You are designing a database for KW Humane Society. The result is the following set of relations where the type of each relations attribute is given following the attribute (e.g., ID: integer):

Animals(ID: integer, Name: string, PrevOwner: string, DateAdmitted: date, Type: string)

Adopter(SIN: integer, Name: string, Address: string, OtherAnimals: integer)

Adoption(AnimalID: integer, SIN: integer, AdoptDate: date, chipNo: integer) where

The primary keys are underlined. Animals stores information about the animals currently at the Humane Society. Each is given an ID, and their names together with the SIN of their previous owners (attribute PrevOwner), and their date of admission is recorded. Type refers to the type of animal (dog, cat, etc). Adopter is the relation that holds information about animal adopters. The attributes are self-descriptive, except OtherAnimals which records the number of other animals that the adopter currently has at home. AnimalID in Adoption refers to the ID of Animals. Similarly, SIN in Adoption refers to the SIN of Adopter. Attribute chipNo stores the number on the microchip that is implanted on the animal for tracking. Owner in Animals refers to the SIN of Adopter.

Formulate the following queries in SQL;

- a) Retrieve the total number of dogs that were brought to the Humane Society on 18 April 2000. **5marks**



- b) List the name of the adopter who has adopted every type of animal. **5marks**
- c) Define Entity, Entity type, and Entity set. **3marks**
- d) What is Data Warehousing? **2marks**

**SECTION C: MOBILE OPERATING SYSTEM (25marks)**

- 1) What is the proper way of setting up an Android-powered device for app development? **5marks**
- 2) What composes a typical Android application project? **5marks**
- 3) When is the best time to kill a foreground activity? **5marks**
- 4) Do all mobile phones support the latest Android operating system? **5marks**
- 5) What is an action in mobile computing OS? **5marks**

**SECTION D: DATABASE ADMINISTRATION WITH MYSQL: 25MARKS**

- 1) What are the commands used in DCL? **5marks**
- 2) What is a system database and what is a user database? **5marks**
- 3) What is Replication? **5marks**
- 4) If you are given access to a SQL Server, how do you find if the SQL Instance is a named instance or a default instance? **5marks**
- 5) What are the different authentication modes in SQL Server and how can you change authentication mode? **5marks**



**National Exam of Higher National Diploma 2023 Session**

**Speciality/Option: SOFTWARE ENGINEERING (SWE)**

**Paper : Computer Technology**

**Duration : 4 hours**

**Credit : 6**

**ANSWER ALL QUESTIONS**

**SECTION A: MCQS (40marks)**

**Each question carries One (0 1) mark.**

- 1- Which of the following programming language is whitespace insensitive?
  - a. HTML
  - b. CSS
  - c. PHP
  - d. JAVASCRIPT
- 2- How many layers does OSI Reference Model has?
  - a. 4
  - b. 5
  - c. 6
  - d. 7
- 3- DHCP is the abbreviation of
  - a. Dynamic Host Control Protocol
  - b. Dynamic Host Configuration Protocol
  - c. Dynamic Hyper Control Protocol
  - d. Dynamic Hyper Configuration Protocol
- 4- An example of Private IP address is:
  - a. 10.2.5.8.9
  - b. 195.25.87.96
  - c. 204.14.88.96
  - d. 173.29.178.66



5- An example of Public IP address is:

- a. 10.2.5.8.9
- b. 192.168.87.96
- c. 204.14.88.96
- d. 172.29.178.66

6- SNMP protocol means:

- a. Simple Network Mail Protocol
- b. System Network Management Protocol
- c. Simple Network Management Protocol
- d. All above

7- The language used in application programs to request data from the DBMS is referred to as the:

- a. DML
- b. DDL
- c. VDL
- d. SDL

8- `int z,x=5,y=-10,a=4,b=2; z = x++ --y * b / a;`

What number will z in the sample code above contain?

- a.
- b. 5
- c. 6
- d. 10
- e. 11

9- With every use of a memory allocation function, what function should be used to release allocated memory which is no longer needed?

- a. dropmem()
- b. dealloc()
- c. release()
- d. free()

10- What function will read a specified number of elements from a file?

- a. fileread()
- b. readfile()
- c. fread()
- d. gets().



- 11- What is a difference between a declaration and a definition of a variable?
- a. Both can occur multiple times, but a declaration must occur first.
  - b. There is no difference between them.
  - c. A definition occurs once, but a declaration may occur many times.
  - d. A declaration occurs once, but a definition may occur many times.
- 12- What is a proper method of opening a file for writing as binary file?
- a. `FILE *f = fwrite( —test.binl, —bl );`
  - b. `FILE *f = fopenb( —test.binl, —wl );`
  - c. `FILE *f = fopen( —test.binl, —wbl );`
  - d. `FILE *f = fopen( —test.binl, —bwl );` [Ans]
- 13- A collection of hyperlinked documents on the internet forms the:
- a. World Wide Web (WWW)
  - b. E-mail system
  - c. Mailing list
  - d. Hypertext markup language
- 14- A proxy server is used as the computer?
- a. with external access
  - b. acting as a backup
  - c. performing file handling
  - d. accessing user permissions
- 15- Which one of the following would breach the integrity of a system?
- a. Looking the room to prevent theft
  - b. Full access rights for all users
  - c. Fitting the system with an anti-theft device
  - d. Protecting the device against willful or accidental damage
- 16- Which software prevents the external access to a system?
- a. Firewall
  - b. Gateway
  - c. Router
  - d. Virus checker
- 17- Where is linear searching used?
- a. When the list has only a few elements
  - b. When performing a single search in an unordered list
  - c. Used all the time
  - d. When the list has only a few elements and When performing a single search in an unordered list



18- What is the best case for linear search?

- a.  $O(n \log n)$
- b.  $O(\log n)$
- c.  $O(n)$
- d.  $O(1)$

19- Which of the following is a disadvantage of linear search?

- a. Requires more space
- b. Greater time complexities compared to other searching algorithms
- c. Not easy to understand
- d. Not easy to implement

20- Which of the following is a disadvantage of linear search?

- a. Requires more space
- b. Greater time complexities compared to other searching algorithms
- c. Not easy to understand
- d. Not easy to implement

21- Which of the following concepts make extensive use of arrays?

- a. Binary trees
- b. Scheduling of processes
- c. Caching
- d. Spatial locality

22- Assuming int is of 4bytes, what is the size of `int arr[15];`?

- a. 15
- b. 19
- c. 11
- d. 60

23- The data structure required for Breadth First Traversal on a graph is?

- a. Stack
- b. Array
- c. Queue
- d. Tree

24- Cartesian product in relational algebra is

- a. Binary operator
- b. Unary operator
- c. Ternary operator
- d. Not defined

25- The view of total database content is:

- a. Conceptual view
- b. Internal view
- c. Physical view
- d. External view

26- Architecture of a database can be viewed as:

- a. One level
- b. Two levels
- c. Three levels
- d. Four levels

27- Database schema is written in:

- a. DML
- b. HCL
- c. DDL
- d. DCL

28- Which technique is used for data protection?

- a. Data piracy
- b. Authentication
- c. Encryption
- d. None of these

29- Coaxial cables used in \_\_\_\_\_ networks

- a. Telephone
- b. Cable TV
- c. Both A and B
- d. NONE OF these

30- Optical Fiber transmit \_\_\_\_\_.

- a. light signal
- b. radio signal
- c. electrical signal
- d. Both A and B

31- HTTP port number used is \_\_\_\_\_.

- a. 43
- b. 441
- c. 443
- d. 449



32- E. In the handshake protocol which is the message type first sent between client and server?

- a. server\_hello
- b. client\_hello
- c. hello\_request
- d. certificate\_request

33- Inside which HTML element do we put the JavaScript?

- a. <js>
- b. <javascript>
- c. <scripting>
- d. <script>

34- Which one is not a programming language?

- a. HTML
- b. CSS
- c. JavaScript
- d. PHP

35- Which one is a method of inserting a style sheet?

- a. Inline style sheet
- b. External style sheet
- c. Internal style sheet
- d. All above

36- An example of a variable declaration in JavaScript:

- a. \$a;
- b. Var a;
- c. Variable a;
- d. None above

37- An example of a variable declaration in PHP:

- a. \$a;
- b. Var a;
- c. Variable a;
- d. None above

38- The RSA signature uses which hash algorithm?

- a. MD5
- b. SHA-1
- c. MD5 and SHA-1
- d. None of the mentioned.



39- Choose the function of a router

- ✓ a. changing the data from one format to another
- b. error detection in data
- c. send the packet to the uplinks
- d. None of the above

40- Which utility used, If you want to find the number of routers between a source and destination.

- a. route
- b. Traceroute
- c. Ifconfig
- d. Ipconfig

## **SECTION B: QUESTIONS/ANSWERS (60marks)**

### **A. Programming (20marks)**

- 1) What is a Programming Paradigm? **2 marks**
- 2) Is it true that Operating Systems like Windows, Linux and UNIX are written in C? **2 marks**
- 3) What do you mean by scope of a variable? What are the different types of scopes that a variable can have? **2 marks**
- 4) Which of the following statement is a declaration and which is a definition?  
`extern int i ; int j = 0 ;` **2 marks**
- 5) What are the differences between a declaration and a definition? **2 marks**
- 6) Is it true that a global variable may have several declarations, but only one definition?  
[Yes/No] **2 marks**
- 7) Is it true that a function may have several declarations, but only one definition?  
[Yes/No] **2 marks**
- 8) When we mention the prototype of a function are we defining the function or declaring it? **2 marks**
- 9) If you are to share the variables or functions across several source files, how would you ensure that all definitions and declarations are consistent? **2 marks**
- 10) Global variables are available to all functions. Does there exist a mechanism by way of which I can make it available to some and not to others? **2 marks**



**B. Database (15 Marks)**

- 1) Differentiate Table Scan from Index Scan. **4 marks**
- 2) List some commands of DDL. **4 marks**
- 3) Why is group-clause used? **3 marks**
- 4) What are the various kinds of interactions catered by DBMS? **4 marks**

**C. Networking (15 marks)**

- 1) What is OSI and what role does it play in computer networks? **3 marks**
- 2) What is the purpose of cables being shielded and having twisted pairs? **4 marks**
- 3) One way of securing a network is through the use of passwords. What can be considered as good passwords? **4 marks**
- 4) Name the two data link sublayers and their purposes. **4 marks**

**D. Algorithms and Data Structure (10 Marks)**

- 1) How to find whether the linked list has a loop? **(3mks)**
- 2) Differentiate between best case scenario and worst case scenario of an algorithm. **(4mks)**
- 3) Describe the two principal operation that can be performed on a Stack. **(3mks)**



**National Exam of Higher National Diploma 2023 Session**

**Spécialty/option : All Specialties**

**Paper: English Language**

**Duration : 2 hours**

**Credit : 1/2**

**SECTION A: READING COMPREHENSION (20 mks)**

Read the following passage carefully and answer the questions that follow.

Defeat of the mind is the surest evidence of defeat of the man himself. In our nation today, the most common and overused statement is “the youths of today are the leaders of tomorrow”. But how true is this statement? It is no news that youths in our nation have already been defeated in their minds by the notion that leadership is only for the old and red headed. This is a false perspective from which to look at things. A famous public speaker Denzel Washington said “Without commitment, you will never start. But more importantly, without consistency, you will never finish”. Let us analyse this in our 21<sup>st</sup> Century context.

Commitment is a virtue that has proven to guide and govern the 21<sup>st</sup> Century world. It is the first step to progress and inherent success. President Donald J. Trump was a real trump card in the most recent Presidential elections in the United States of America. For him, there was hardly ever any sign of positive progress; still, he proved his person and won the position. Remember that even US citizens were taken aback when the results were declared. He was committed to the cause and so he started it, but had he not been consistent, he would have lost without a doubt even before the end.

Look at George Weah, President of the Republic of Liberia and former soccer player in Cameroon. I believe it was the scariest part of his life when he decided to retire from international soccer to dare into politics. Some anonymous reports even hold that he left because he wasn't fully accomplished where he was at the time. If this is true, then I can start imagining what went through his mind. Thoughts like there is more potential in me than football can explore. When this crossed his mind, he decided to make the difference in his country. Even



after many failed attempts against his predecessor, when the time came, his capable hands were the only available ones for the job.

At this point, I must say just as great and successful figures do that “True success only comes when opportunity meets preparation” (anonymous). President George Weah was committed, **consistent** and prepared; when the opportunity came, he succeeded.

Did you know that Davido’s father is Daliko Dangote; one of the world’s richest black men? If you knew, I didn’t. When I found out this, I asked myself a **multi-million dollar question**; what will a typical Cameroonian youth do if put in Davido’s shoes? He will probably sit back, relax and spend money **outrageously**. Interestingly, he will do this with no sense of purpose or positive drive. Yet, you want to lead tomorrow. .

I stand here today, as a voice of inspiration and a beacon of hope to say that it is not over yet. If only we could value schools more than nightclubs, books more than clothes and education more than fashion, then we can make for ourselves a better tomorrow. Remember that **rock bottom** can also serve as a solid foundation on which to build the mansion of our desired society. Young Cameroonians; stop existing and start living.

### Questions

1. What type of write up (essay) is employed above? (1mk)
2. List the other essay types that you know in English writing. (2mks)
3. What is the paragraph format used in the passage you just read? (1mk)
4. Carefully identify and write out the take home message above. (2mks)
5. Write out the bolded words and provide the most suitable antonyms (opposites) in your answer sheets. (2mks)
6. “True success only comes when opportunity meets preparation” (anonymous). What is your interpretation of this statement? (2mks)
7. Following public speaking and writing, what name will you give to this text/passage?(1mk)
  - a. Speech
  - b. Talk
  - c. Discourse
  - d. Campaign
8. Which is the dominant tense used in the passage above?(1mk)
  - a. The future imperfect
  - b. The perfect present



c. The present simple

d. The past perfect

9. According to the passage, state two ways that Cameroonian youth are encouraged. (2mks)
10. Paragraph 5 uses 2 question types which have been underlined. Identify the two question types used in that paragraph. (2mks)
11. Apart from the two question types used in question 10 above, list the other question types that you know. (2mks)
12. Write down the names of 4 role models who appear in the passage. (2mks)

**SECTION B: GRAMMAR AND VOCABULARY (30 marks)**

**GRAMMAR (10 marks)**

**I. Fill in the blanks with one of the adjectives in brackets (5 marks)**

1. He is the (strong) boy in the whole school. (1mk)
2. Of the two sisters, Mary is the (beautiful) (1mk)
3. Ann is the (young) of four sisters. (1mk)
4. David is (bad) than his brother (1mk)
5. Which of the two girls is (tall)? (1mk)

**II. Match the sentence parts in column A with the appropriate ones in column B. Write the correct sentence in the space provided (5 marks).**

|   | Column A                                 | Column B                                 |
|---|------------------------------------------|------------------------------------------|
| 1 | Bantar went to the market                | a) If I worked had                       |
| 2 | The teacher gave his students more marks | b) if I went to school.                  |
| 3 | Sarah went to the hospital               | c) to get some food stuff.               |
| 4 | I would have passed my exams             | d) for check-up                          |
| 5 | I would have had money                   | e) in order for them to pass their exams |

- 1.....(1mk)
- 2.....(1mk)
- 3.....(1mk)
- 4.....(1mk)
- 5.....(1mk)



## VOCABULARY (20 marks)

### I. Complete the following sentences with suitable words (5marks)

- Gabriel, how would you refuse to help your mother with the.....(1mk)
- All names of places begin with.....letters. (1mk)
- I do not eat before going to church. I take some..... in the morning and then I eat dinner at home with my family. (1mk)
- Blood shortage in the body is also referred to as.....(1mk)
- Aids is real, .....? (1mk)

### II. Find the appropriate part in column B that fits with that of column A. Write out the full expression in the spaces provided. (6 marks)

|   | Column A                  | Column B            |
|---|---------------------------|---------------------|
| 1 | It is often difficult     | a) Twice shy        |
| 2 | To take care              | b) nor pink color   |
| 3 | The most intelligent girl | c) To say "sorry"   |
| 4 | It is disrespectful       | d) of someone       |
| 5 | Once beaten               | e) in school        |
| 6 | It is neither rose        | f) to scorn someone |

- .....(1mk)
- .....(1mk)
- .....(1mk)
- .....(1mk)
- .....(1mk)
- .....(1mk)

### III. Link the following pairs of sentences using the words supplied in the brackets.

(5 marks)

- The man is shopping in the market. The man is my brother (who) (1mk)
- The car was stolen. The car has been found (which). (1mk)
- Ngong is not sure. He shall go to the party (that). (1mk)
- Shella and Cynthia boarded a bike. There were no taxis (because) (1mk)
- Epie delayed. It was raining (since). (1mk)



*Circle the word in which the underlined word is pronounced differently (4marks)*

1. Pound, stout, though, mouth. (1mk)
2. Tear, near, gear, tease (1mk)
3. Kneel, peel, heel, peer (1mk)
4. Suit, put, shoot, shut (1mk)

**SECTION C: SPOKEN ENGLISH (20 marks)**

**I. Complete the sentences with the correct word chosen from the brackets (5 marks)**

1. The President.....his wife, is going abroad for a diplomatic concern. (and, with, for) (1mk)
2. The Manager and.....did all the work yesterday. (I, me, ourselves) (1mk)
3. Look at that couple, they truly love.....(each other, themselves, one another) (1mk)
4. The Director was very angry.....the problem that occurred. (with, at, for) (1mk)
5. Tell me what you are good.....doing. (wit, in, at) (1mk)

**II. Complete the sentences with suitable words. (5 marks)**

1. He is too selfish, he does not really care.....other people around him. (1mk)
2. They all sat.....the table to eat. (1mk)
3. You have all the credentials needed for the job, why don't you apply.....it (1mk)
4. The boss congratulated me.....my success in the assignment given to me. (1mk)
5. He spends all his time lying.....bed. (1mk)

**III. Circle the odd sound. ( 10 marks)**

1. Sure, send, shy (1mk)
2. Unveil, university, youth (1mk)
3. Laugh, phone, though (1mk)
4. Exist, exam, excellent (1mk)
5. Go, no, women (1mk)
6. Player, mayor, major (1mk)
7. Girls, zeal, since (1mk)
8. Power, shown, hour (1mk)



9. Great, grateful, meal (1mk)

10. Thin, thick, this (1mk)

**SECTION D: WRITING COMPETENCE (30 marks)**

*Write an essay on one of the following topics (between 250 to 300 words).*

1. On the National Youth day, you have been requested by the Management of your school in the presence of the Governor to make a speech on the importance of language to professional students. As the president of the Students Union of your school, write your speech.
2. Insecurity at workplaces
3. You are the secretary of the staff of the regional hospital in Bamenda. Write minutes of the meeting of this body.



RÉPUBLIQUE DU CAMEROUN  
Paix-Travail-Patrie

MINISTÈRE DE L'ENSEIGNEMENT SUPÉRIEUR

COMMISSION NATIONALE D'ORGANISATION DE L'EXAMEN  
HIGHER NATIONAL DIPLOMA EXAM (HND)

REPUBLIC OF CAMEROON  
Peace-Work-Fatherland

MINISTRY OF HIGHER EDUCATION

NATIONAL COMMISSION FOR THE ORGANISATION OF  
HIGHER NATIONAL DIPLOMA (HND) EXAM

**National Exam of Higher National Diploma 2023 Session**

**Spécialty/option :** All Specialties

**Paper:** French Language

**Duration :** 2 hours

**Credit :** 1/2

**Instructions:** Cochez la bonne réponse.

**A. La Langue**

**1 – Grammaire, Conjugaison et Vocabulaire.**

1. \_\_\_\_\_ malade qui est couchée est très souffrante.  
A) Le            B) La            C) Une            D) Un
2. Mon frère a \_\_\_\_\_ meubles dans son salon.  
A) Des jolis            B) De joli            C) De jolis            D) Des joli
3. Qui va \_\_\_\_\_ va sûrement.  
A) Lentement            B) gentiment            C) régulièrement            D) subitement
4. Helene a écrit une lettre à sa sœur. Elle \_\_\_\_\_ écrite.  
A) Le la            B) La lui            C) Le lui            D) Lui la
5. Le professeur va au cours \_\_\_\_\_ vélo.  
A) Sure            B) En            C) Par            D) à
6. Buvez-vous du vin? Oui, nous \_\_\_\_\_ buvons.  
A) Le            B) La            C) y            D) en
7. Le père ne mange pas \_\_\_\_\_ viande.  
A) De la            B) Du            C) De            D) La
8. Je te donne ces conseils \_\_\_\_\_ tu deviennes sage.  
A) Parce que            B) Pour que            C) Puisque            D) Alors que
9. Mon professeur parle à mes parents. IL \_\_\_\_\_ parle.  
A) Les            B) Leurs            C) Leur            D) à eux



10. Les gens qui vivent en ville sont des \_\_\_\_\_.

- A) Citoyens B) Citadelles C) Citoyennes D) Citadins

11. J'ai présenté les fleurs à ma sœur.

- A) Je la lui ai présentée B) Je les lui ai présentées  
C) Je la les ai présentées D) Je les ai présentées.

12. A la forme négative, la phrase "t'es-tu lavé" s'écrira:

- A) Ne t'es lave-tu pas? B) N'est-pas tu t'es lave?  
C) N'est-ce pas tu t'es lave? D) Ne t'es-tu pas lavé?

13. \_\_\_\_\_ temps après la visite, les visiteurs ont repris le chemin du retour.

- A) Quelque B) Quelques C) Quel que D) Quels que

14. IL y a \_\_\_\_\_ élèves en classe aujourd'hui.

- A) Quatre-vingt B) Quatre vingts C) Quatres vingt D) Quatresvingts

15. La troupe compte \_\_\_\_\_ soldats.

- A) Deux milles deux cents B) Deux mille deux cent C) Deux mille deux cents  
D) Deux milles deux cent

16. Voicile \_\_\_\_\_ qui a opéré ma sœur.

- A) Chirurgienne B) Garde-malade C) Boucher D) Chirurgien

17. \_\_\_\_\_ les enfants de ce monsieur sont parties en congé.

- A) Tous B) Toutes C) Tout D) Toute

**B. Traduction, expressions, voix active et passive.**

18. "Rien ne sert de courir. Il faut partir a point" veut dire qu'il

- A) Faut se précipiter juste un peu B) N'est pas bon de se précipiter  
C) Ne faut jamais courir D) Faut vite faire ce qu'on a à faire.

19. "Paul ne mange pas à sa faim", veut dire:

- A) Qu'il ne mange pas quand il a faim B) Qu'il mange quand il a faim  
C) Qu'il ne mange pas ce qu'il veut. D) Qu'il ne mange pas suffisamment.

20. "Un ministre sans porte-feuille" veut dire:

- A) Qui n'a pas de porte-monnaie B) Qui n'est pas à la tête d'un ministère  
C) Qui ne porte pas les feuilles D) Qui n'a pas de feuilles dans son bureau.



21. "My father is a teacher at the university" can be translated as:

- A) Mon père est un enseignant à l'université
- B) Mon père est l'enseignant de l'université
- C) Mon père c'est l'enseignant de l'université
- D) Mon père est enseignant à l'université.

22. Ten years ago Cameroon was a paradise.

- A) Dix ans passés le Cameroun était un paradis.
- B) Il y a dix ans, le Cameroun était un paradis
- C) Le Cameroun était un paradis depuis dix ans
- D) Il y a dix ans, le Cameroun est un paradis.

23. A la forme passive la phrase "Paul dessine un éléphant".

- A) Un éléphant est dessiné par Paul
- B) Un éléphant a été dessiné par Paul
- C) Un éléphant sera dessiné par Paul
- D) Un éléphant était dessiné par Paul.

24. "At the university, any student is entitled to a scholarship".

- A) A l'université, n'importe quel étudiant a droit à la bourse.
- B) A l'université, même quel étudiant a droit à la bourse
- C) A l'université, quelques étudiants ont droit à la bourse
- D) A l'université, certains étudiants ont droit à la bourse.

25. "Un but sera marqué par Eto'o Fils". A la voix active s'écrit :

- A) Eto'o Fils marquera un but.
- B) Eto'o Fils a marqué un but.
- C) Eto'o Fils marquerait un but.
- D) Eto'o Fils marque un but.

26. "Marie a préparé la nourriture" à la voix passive s'écrit:

- A) La nourriture avait été préparée par Marie.
- B) La nourriture était préparée par Marie
- C) La nourriture a été préparée par Marie.
- D) La nourriture a été préparé par Marie.

**C. Sport – Culture générale – Santé – Politique Economie**

27. La première université créée au Cameroun c'est celle de ?

- A) Ngaoundéré
- B) Bamenda
- C) Yaoundé 1
- D) Yaoundé 2

28. Le Chef du gouvernement c'est \_\_\_\_\_.

- A) AtangaNji Paul
- B) DionNguteh
- C) Jacques FameNdongo
- D) Ferdinand Ngo Ngo.



29. Le congrès de Bamenda qui a vu la naissance du RDPC a eu lieu en :  
 A) 1982      B) 1990      C) 1985      D) 1992
30. Le Cameroun est devenu la République du Cameroun en :  
 A) 1972      B) 1985      C) 1984      D) 1982
31. La Devise du Cameroun c'est  
 A) Patrie, Travail-Paix      B) Travail, Patrie, Paix  
 C) Paix - Travail - Patrie      D) Paix - Patrie - Travail.
32. Les couleurs du drapeau du Cameroun sont :  
 A) Rouge - Vert - Jaune      B) Vert - Jaune - Rouge  
 C) Jaune - Rouge - Vert      D) Vert - Rouge - Jaune
33. Les Lions Indomptables à qui l'Etat leur offert des maisons ont participé à la coupe du monde en :  
 A) 1994      B) 1990      C) 2006      D) 1998
34. Le ministre de l'enseignement supérieur s'appelle :  
 A) Nalova Lyonga      B) Jacques Fame Ndongo      C) Beti Assomo      D) Gregoire Owona
35. En quelle année fut l'accession du président Paul Biya à la Magistrature Suprême du Cameroun?  
 A) En 1985      B) En 1982      C) En 1984      D) En 1990
36. Le port du cache-nez est exigé pour :  
 A) Le paludisme      B) Le SIDA      C) Le COVID - 19      D) La toux
37. Mettre quelqu'un en quarantaine c'est :  
 A) L'isoler des autres      B) Le garder pendant quarante jour  
 C) Le chasser des autres      D) L'empêcher de fuir.
38. Les jeux universitaires se déroulent \_\_\_\_\_.  
 A) Chaque année      B) Tous les trois ans  
 C) Tous les deux ans      D) Tous les quatre ans
39. Les 11 février, c'est \_\_\_\_\_.  
 A) La fête nationale      B) La fête de la jeunesse  
 C) La fête de l'indépendance      D) La fête de la Réunification
40. Le Président du parti Mouvement pour la Renaissance du Cameroun (MRC) c'est :  
 A) Cabral Libi      B) Kawala Edith      C) Maurice Kamto      D) Jean Jacques Ekindi



## **SECTION II: COMPREHENSION DE TEXTE (30 pts: 30%)**

**Lisez le texte suivant et répondez aux questions qui suivent**

Camerounaise, Camerounais, mes chers compatriotes, au cours de l'année qui s'achève, le Cameroun, comme la plupart des pays a évolué dans un contexte encore fortement marqué par la crise économique et financière mondiale. Il y a un an, alors que nous commençons à en ressentir les premiers effets, je vous disais qu'à première vue l'impact de la crise pourrait, en ce qui nous concerne, être relativement limité. En général, ces hypothèses ont été vérifiées. Notre secteur financier a assez bien résisté. En revanche, nos exportations de bois, d'aluminium, de coton et du caoutchouc ont régressé, recul aggravé par la baisse des coûts des matières premières et des crises du secteur d'électricité. Ce ralentissement de notre activité économique n'a pas été sans conséquences. Les recettes fiscales et douanières ont diminué et l'emploi en a souffert. Comme il était prévisible, les investissements en provenance de l'extérieur ont été différés et le crédit s'est resserré.

Au total notre économie a fléchi, c'est vrai mais elle n'a pas été ébranlée dans ses fondations, comme dans certains pays et non les moindres. Certaines activités productrices notamment dans l'agriculture ont même progressé. Il n'en reste pas de moins que le taux de croissance de notre produit intérieur Brut a marqué le pas. En 2009, il sera en recul par rapport à l'année précédant. Dans une telle situation, que convenait-il de faire? Ainsi que j'ai eu l'occasion de le dire à plusieurs reprises, il fallait garder le cap, agir d'urgence au profit de la population la plus exposée au ralentissement de l'économie et anticiper la sortie de crise. C'est ce que nous avons fait en exerçant une action modératrice sur les prix des matières de premières nécessités en soutenant la consommation et en reprenant de façon significative les recrutements dans la fonction publique. Cela n'a pas été facile, car nous devons, en raison de nos recettes, maintenir une discipline budgétaire aussi stricte que possible pour ne pas compromettre l'avenir.

### **QUESTIONS (10pts) :**

1. Donner un titre approprié à ce texte après avoir identifié son auteur. (1pt)
2. En quelle année et à quelle occasion a-t-il prononcé ce discours ? (2pts)
3. Quel secteur de l'économie a pu lutter la crise économique ? (1pt)
4. Citez deux mesures de sortie des crises prises par le gouvernement. (2pts)
5. Que veut dire l'expression : notre économie a fléchi? (1pt)
6. Quelles sont les conséquences du ralentissement de l'activité économique? (2pts)
7. Que faire pour ne pas compromettre l'avenir? (1pt)

**Former des verbes à partir des noms suivants (5pts)**

Consommation – Ralentissement – Investissement – Progrès – Production.



**Accordez correctement les participes passés des verbes entre parenthèses. (5pts)**

1. Le professeur nous a (donner) une leçon à apprendre demain.
2. Savez-vous par qui sont (interroger) les suspects?
3. Ils applaudissaient à tout rompre, enthousiasmé par la pièce qu'ils avaient (voir).
4. La déléguée que les étudiants ont élue a (rendre) compte du rapport.
5. L'expert que mes parent ont consulté leur a (monter) les défauts du plan.

**Mettez les verbes entre parenthèses aux temps indiqués (5pts)**

1. La Croix-Rouge (envoyer) des médecins distribués des médicaments. (futur).
2. Nous (tenir) – rapidement conseil et (recevoir) – la délégation. (passé composé).
3. (Se lever), (faire) la lessive, (manger) vite, si non, tu seras en retard à l'école. (Impératif)

**SECTION III: Essay (30 pts: 30%)**

**Write an essay of not more than 200 words on one of the following topics.**

1. Le rêve de tout jeune Camerounais diplômé est de toujours partir. Pour vous, l'émigration est-elle le seul moyen de réussir dans la vie?
2. Si vous étiez élu Président de la République, quelles mesures prendriez-vous pour lutter contre la corruption à tous les niveaux?



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.....  
MINISTÈRE DE L'ENSEIGNEMENT SUPÉRIEUR

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HIGHER NATIONAL DIPLOMA EXAM (HND)

REPUBLIC OF CAMEROON  
Peace-Work-Fatherland

.....  
MINISTRY OF HIGHER EDUCATION

.....  
NATIONAL COMMISSION FOR THE ORGANISATION OF  
HIGHER NATIONAL DIPLOMA EXAM (HND) EXAM

**National Exam of Higher National Diploma 2023**

**Spécialty/option : All specialities**

**Paper : Enterprise Creation and Entrepreneurship**

**Duration : 2 HOURS**

**Credit :1**

**SECTION 1: ENTREPRENEURSHIP (34marks)**

1. Define the following (10mks)
  - a) Franchise
  - b) Venture capital
  - c) Business venture
  - d) Innovation
2. What is an "executive summary"? Why the executive summary is often called the most important part of a business plan? (14mks)
3. Explain the life cycle of a small business to an entrepreneur (10mks)

**SECTION 2: GENERAL ECONOMICS (33marks)**

4. The individual demand and supply function of cell phones in a market is given by:  
 $Q_d = 2000 - 2P$   
 $Q_s = 400 + 2P$  where  $Q_d$ ...individual demand,  $Q_s$  ...individual supply,  $P$ ...is the price of cell phones
  - a) Determine the equilibrium price and quantity. (5mks)
  - b) If the government guarantee a price of 600 FRS, calculate the surplus or the shortage in the market and state the effects. (5mks)
  - c) Calculate the price elasticity of demand if price were to increase by 100 FRS from the equilibrium. (4mks)
5. State and explain five (5) qualities of money. (10mks)
6. State five (5) effects of the African cup of nations 2021 to the economic growth of Cameroon. (10mks)



**SECTION 3: COMPANY LAW (33MARKS)**

7. Define an Article of Association and briefly explain five of its component. **(13mks)**
8. Examine five types of companies created under the OHADA Uniform Act. **(20mks)**



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### National Exam of Higher National Diploma 2023

Spécialty/option : FOT, APT, CPT, APA, AGE, AGP, CET, UPL, BST, CAR, JOC, RCE, MEM, CHM, CHP, ELT, EPS, ACR, QOP, PSE, PLO, ASM, PMA, HUR, ACC, MTS, BFI, INT, BFP, FCT, TAT, HMC, DCT, EMA, SPE, NWS, SWE, CSN, HWM, ICA, CWD, EDM, NGO, AND, OPT, BCR, SPW, BMW, CUT, ISM, LGA, PCIL, TOP, AMS, APG, SOW, FSD, DBM, AFI, EIA, MET, MEC, ELN, BBA

Paper : LAW AND CITIZENSHIP EDUCATION

Duration : 2 HRS

Credit : 2

Instructions: Answer all questions. Logical presentation, grammar, neat work and good handwriting counts.

#### SECTION A: INTRODUCTION TO LAW AND FUNDAMENTAL RIGHTS. (20 marks)

1. "If law was not created it could have been invented". Discuss.

#### SECTION B: CIVICS, ETHICS, MORAL AND CITIZENSHIP EDUCATION. (20 marks)

2. a) What are Human rights? Explain any four features of human rights. (10 marks)

- b) Discuss four rights and duties of Cameroonian Citizens. (10 marks)

- To work - To education  
- To take - To belong to a religion

#### SECTION C: LABOUR LAW (20 marks)

3. Briefly discuss the rights of an employee in a labour contract.

#### SECTION D : BUSINESS LAW (20 marks)

4. Explain any five advantages of arbitration over litigation or court action.

#### SECTION E: COMPANY LAW (20 marks)

5. Examine the discrepancies that exist between Public Limited Companies and Private Limited Companies as provided for by the OHADA Law.